

Bonus questions: Random Variables and Probability Distributions

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B1. The inverse-transform method. Let F be any distribution function. For $0 < u < 1$, define its generalized inverse (or quantile function) by

$$Q(u) = \inf\{x \in \mathbb{R} : F(x) \geq u\}.$$

Let U be uniformly distributed on $(0, 1)$, and put $X = Q(U)$.

- (a) Using monotonicity and right continuity of F , show that for every $x \in \mathbb{R}$ and $0 < u < 1$,

$$Q(u) \leq x \iff u \leq F(x).$$

- (b) Hence prove that

$$\mathbb{P}(X \leq x) = F(x)$$

for every $x \in \mathbb{R}$. Thus every distribution function can be realized by transforming a uniform random variable.

- (c) Suppose

$$F(x) = \begin{cases} 0, & x < 0, \\ 1 - e^{-\lambda x}, & x \geq 0, \end{cases} \quad \lambda > 0.$$

Find $Q(u)$ explicitly. Give a formula for generating an exponential random variable from a uniform random variable.

- (d) Let $0 < p < 1$ and let F be the distribution function of a Bernoulli random variable with

$$\mathbb{P}(X = 1) = p, \quad \mathbb{P}(X = 0) = 1 - p.$$

Find $Q(u)$ explicitly. Notice that the same construction works even though F has a jump.

B2. A continuous distribution with no density. This question constructs the *Cantor distribution*. Consider an infinite sequence of fair coin tosses. Write

$$\omega = (\omega_1, \omega_2, \dots), \quad \omega_k \in \{0, 1\},$$

and assume that every specified string of the first n tosses has probability 2^{-n} . Define

$$X(\omega) = \sum_{k=1}^{\infty} \frac{2\omega_k}{3^k}.$$

The series converges and takes values in $[0, 1]$.

Recall the middle-thirds construction of the Cantor set C : begin with $[0, 1]$, remove the open middle third, then remove the open middle third of each remaining interval, and continue indefinitely.

- (a) Show that the ternary expansion of X uses only the digits 0 and 2. Deduce that

$$\mathbb{P}(X \in C) = 1.$$

- (b) Fix $x \in [0, 1]$. Show that the event $\{X = x\}$, if nonempty, determines the first n coin tosses for every n . Hence prove that

$$\mathbb{P}(X = x) \leq 2^{-n}$$

for every n , and conclude that $\mathbb{P}(X = x) = 0$.

- (c) Using $\mathbb{P}(X = x) = F(x) - F(x^-)$, deduce that the distribution function F of X is continuous everywhere.

- (d) After n stages of the Cantor-set construction, there are 2^n intervals, each of length 3^{-n} . Show that their total length is

$$\left(\frac{2}{3}\right)^n,$$

and hence that the Cantor set has length 0.

- (e) Suppose, for contradiction, that X had a probability density f . You may use the standard fact that a random variable having a density assigns probability 0 to every set of length 0. Explain why this contradicts part (a).
- (f) Conclude that X has a continuous distribution function but no probability density. Such a distribution is called *singular continuous*. Why does this show that the words “discrete” and “continuous (with a density)” do not exhaust all random variables?